

Technology trends

Embedded components. SMT components come in various packages. With advancements in technology, the size of many packages has reduced. Most SMT component sizes adhere to industry standards like JEDEC specifications. Use of standards simplifies PCB design as well as manufacture (as pick-and-place machines can use standard feed for SMT components).

Talking of passive SMD components, the 01005 package, or the 0402M, has been in use since around 2002. Though it allows considerable miniaturisation, it is only recently that mobile devices have started using this package. In IC packaging industry, on the other hand, 01005 components are being integrated into ICs since 2002.

Now that 01005 components have found their place onto PCBs, passive component manufacturers are contemplating even smaller components. The next in line are 0201M (0050025) components as well as trends like wafer-level packaging and PCB imprinting. In components, embedded components have already arrived on the scene.

Embedding the components saves on the PCB space. This adds room for additional features on the PCB and also lowers the PCB size, resulting in cost savings. It can also make devices like smartphones thinner.

Coming to IC packaging, quad-flat packages are giving way to ball grid array (BGA). In BGA package, pads underneath the package allow contact with the PCB. Placing the pads underneath the package allows more room, overcoming some of the problems of quad flat packs that require very thin leads. The ball spacing on BGAs is typically 1.27 mm.

Marking on SMD resistors going to disappear! Deciphering those SMD resistor codes has been always a hard task. But things are soon going to change for the better. Passive component provider YAGEO is removing marking of RC/AC 0603, 0805, 1206 SMD chip resistors. This implies that chip resistors are going the chip capacitors way—no marking on the top of the component. This step is aimed to reduce unnecessary chemical usages for environmental protections.

To identify the resistance value of the resistor, every reel will be labeled with all the relevant data. Everything else like performance, features, specifications and manufacturing process of these resistors will also continue to be the same. So removing the marking on top of the resistor itself will not impact the product functionality and reliability. Both types of resistor, with and without marking, will co-exist for some time until all the production lines are changed.

Anticounterfeit packaging. Shortcuts in material development and quality compliance controls can significantly lower operating costs and increase margins. As result, there are vendors that skip these controls and imitate the packaging style of reputed component suppliers to achieve sales with less costs. To counter this problem, suppliers like TDK are offering anti-counterfeit packaging on their select series of MLCCs. This tamperproof packaging style coupled with Web-enabled authentication is seen as the customer's best defense against counterfeit parts.

feels R. Siva Prasad, managing director, Siricom Technology.

The small list of SMD component manufacturers in India includes names like Gujarat Poly-AVX Electronics, Hi-Tech Resistors, Integrated Electronics Technology and Telecommunication Components.

"We have recently started manufacturing bulk metal foil SMD resistors here in India in partnership with Vishay Precision Group. These are ultra-high-stability, very precise

components and we intend to tap the avionics, military and space market," informs Soparkar.

Imports dominate

Talking globally, owing to the fact that the production of chip-based passive components is highly automatic, and therefore requires very little area for machinery, a great many of Japanese companies prefer to produce most of their products within Japan, and turn to China and Southeast Asia

Putting up an SMD manufacturing plant requires a lot of capital. The economy of scale dictates its capacity, which also has to be very large. If a manufacturer were to put up an SMD plant in India, he would have to tap the export market to be viable

for production of small quantities of low-end products only. And the same is true for America, which has little possibility to transfer the production beyond the locality—according to Reportlinker.

"Most of the commercial-grade components are being imported from China, Korea and Taiwan, while professional-grade components are from Europe, Japan and USA. Usually, these components get routed through Singapore as users find it easier to get all their components in kit form from the distributors there," informs Soparkar.

"Components imported from China are poor-quality, while those from countries like Taiwan, Hong Kong and Japan are good in quality," adds Siva Prasad.

A weak rupee is, of course, hitting the import costs and eating into the bottom line. But then there is no other option but to import for the end user.

"We are hit hard by the increase in dollar value but not increasing the prices in short term as we expect the dollar value to stabilise soon," shares Siva Prasad.

EMS impact on component demand

Electronics manufacturing service (EMS) providers are setting up plants mostly in north and south India. But for component distributors in India, this has not translated into increased SMD component sales to these regions.